

Запросы о создании врачей – Кафка получает события о создании:

```
transaction.timeout.ms = 60000
transactional.id = null
value.serializer = class org.springframework.kafka.support.serializer.JsonSerializer

22:13:30.853 [http-nio-8080-exec-3] INFO o.a.k.c.t.i.KafkaMetricsCollector - initializing Kafka metrics collector
22:13:30.861 [http-nio-8080-exec-3] INFO o.a.k.clients.producer.KafkaProducer - [Producer clientId=Medregistry-producer-1] Instantiated an idempotent producer.
22:13:30.870 [http-nio-8080-exec-3] INFO o.a.kafka.common.utils.AppInfoParser - Kafka version: 3.8.1
22:13:30.871 [http-nio-8080-exec-3] INFO o.a.kafka.common.utils.AppInfoParser - Kafka commitId: 70d6ff42debf7e17
22:13:30.871 [http-nio-8080-exec-3] INFO o.a.kafka.common.utils.AppInfoParser - Kafka startTimeMs: 1749323610870
22:13:30.874 [kafka-producer-network-thread | Medregistry-producer-1] INFO org.apache.kafka.clients.Metadata - [Producer clientId=Medregistry-producer-1] Cluster ID: g-DIXY36Sq6TugTEzXW65Q
22:13:30.875 [kafka-producer-network-thread | Medregistry-producer-1] INFO o.a.k.c.p.i.TransactionManager - [Producer clientId=Medregistry-producer-1] ProducerId set to 1001 with epoch 0
22:13:30.918 [org.springframework.kafka.KafkaListenerEndpointContainer#0-0-C-1] INFO r.n.m.s.kafka.DoctorEventConsumer - Получено событие о создании врача: DoctorDTO(id=20, firstName=Иван, lastName=Петров, specialization=Хирург)
Hibernate:
    insert
    into
        doctor
        (deleted, first_name, last_name, specialization)
    values
        (?, ?, ?, ?)
22:15:01.365 [org.springframework.kafka.KafkaListenerEndpointContainer#0-0-C-1] INFO r.n.m.s.kafka.DoctorEventConsumer - Получено событие о создании врача: DoctorDTO(id=21, firstName=Бартоломэо, lastName=Бульдогов, specialization=Экстрасенс)
```

WebSecurityConfig

controller

dto

exceptions

model

repository

security

service

impl

kafka

DoctorEventConsumer

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spring.kafka.template.default-topic=doctor-events

Producer

spring.kafka.producer.key-serializer=org.apache.kafka.common.serialization.StringSerializer

spring.kafka.producer.value-serializer=org.springframework.kafka.support.serializer.JsonSerializer

Consumer

spring.kafka.consumer.key-deserializer=org.apache.kafka.common.serialization.StringDeserializer

spring.kafka.consumer.value-deserializer=org.springframework.kafka.support.serializer.JsonDeserializer

spring.kafka.consumer.properties.spring.json.trusted.packages=*

1

hospitaldb@localhost

hospitaldb 1 of 3

public

tables 7

databasechar

databasechar

doctor

patient

role

run MedregistryApplication x

Console

Actuator

09:51:26.077 [http-nio-8080-exec-6] INFO o.a.k.c.t.i.KafkaMetricsCollector - initializing Kafka metrics collector

09:51:26.085 [http-nio-8080-exec-6] INFO o.a.k.clients.producer.KafkaProducer - [Producer clientId=Medregistry-producer-1] Instantiated an idempotent producer.

09:51:26.095 [http-nio-8080-exec-6] INFO o.a.kafka.common.utils.AppInfoParser - Kafka version: 3.8.1

09:51:26.095 [http-nio-8080-exec-6] INFO o.a.kafka.common.utils.AppInfoParser - Kafka commitId: 70d6ff42debf7e17

09:51:26.095 [http-nio-8080-exec-6] INFO o.a.kafka.common.utils.AppInfoParser - Kafka startTimeMs: 1749624686095

09:51:26.099 [kafka-producer-network-thread | Medregistry-producer-1] INFO org.apache.kafka.clients.Metadata - [Producer clientId=Medregistry-producer-1] Cluster ID: g-DIXY36Sq6TugTEzXW65Q

09:51:26.212 [kafka-producer-network-thread | Medregistry-producer-1] INFO o.a.k.c.p.i.TransactionManager - [Producer clientId=Medregistry-producer-1] ProducerId set to 2000 with epoch 0

09:51:26.245 [org.springframework.kafka.KafkaListenerEndpointContainer#0-0-C-1] INFO r.n.m.s.kafka.DoctorEventConsumer - Получено событие о создании врача: DoctorDTO(id=22, firstName=Захар, lastName=Кузьмин, specialization=Невролог)

Hibernate:

select

d1_0.id,

d1_0.deleted,

d1_0.first_name,

d1_0.last_name,

d1_0.specialization

from

doctor d1_0

where

(

d1_0.deleted=false

)

Удаление врача с ID: 24:

⌕ User

> repository

> security

> service

> impl

> kafka

⌕ DoctorEventConsumer

⌕ DoctorEventProducer

⌕ DoctorService

⌕ PatientService

⌕ UserService

utils

⌕ MedregistryApplication

> resources

> db.changelog

> javadoc

screens

> static.css

> templates

WHERE

ORDER BY

	id	first_name	last_name	specialization	deleted
8	3	Потап	Иванов	лор	• true
9	9	Антон	Чехов	терапевт	• true
10	17	Борис	Борисов	диетолог	false
11	18	Иван	Дляудалёнов	физиолог	• true
12	10	Модест	Юрьев	травматолог	false
13	20	Иван	Петров	Хирург	false
14	21	Бартоломэо	Бульдогов	Экстрасенс	false
15	22	Захар	Кузьмин	Невролог	false
16	23	Кузьма	Захаров	Окулист	false
17	19	Петр	Иванов	Хирург	false
18	26	Анна1	Смирнова	кардиолог	false
19	25	Анна2	Смирнова	кардиолог	false
20	24	Анна3	Смирнова	кардиолог	• true

MedregistryApplication

Console

Actuator

ssl.truststore.password = null
ssl.truststore.type = JKS
transaction.timeout.ms = 60000
transactional.id = null
value.serializer = class org.springframework.kafka.support.serializer.JsonSerializer

12:24:12.091 [http-nio-8080-exec-4] INFO o.a.k.c.t.i.KafkaMetricsCollector - initializing Kafka metrics collector
12:24:12.098 [http-nio-8080-exec-4] INFO o.a.k.clients.producer.KafkaProducer - [Producer clientId=Medregistry-producer-1] Instantiated an idempotent producer.
12:24:12.108 [http-nio-8080-exec-4] INFO o.a.kafka.common.utils.AppInfoParser - Kafka version: 3.8.1
12:24:12.108 [http-nio-8080-exec-4] INFO o.a.kafka.common.utils.AppInfoParser - Kafka commitId: 70d6ff42deb7e17
12:24:12.108 [http-nio-8080-exec-4] INFO o.a.kafka.common.utils.AppInfoParser - Kafka startTimeMs: 1749633852108
12:24:12.111 [kafka-producer-network-thread | Medregistry-producer-1] INFO org.apache.kafka.clients.Metadata - [Producer clientId=Medregistry-producer-1] Cluster ID: g-DIXY3GSq6TugTEzXW65Q
12:24:12.111 [kafka-producer-network-thread | Medregistry-producer-1] INFO o.a.k.c.p.i.TransactionManager - [Producer clientId=Medregistry-producer-1] ProducerId set to 2006 with epoch 0
12:24:12.141 [org.springframework.kafka.KafkaListenerEndpointContainer#0-0-C-1] INFO r.n.m.s.kafka.DoctorEventConsumer - Получено событие об удалении врача с ID: 24

Обновление врача:

User

repository

security

service

impl

kafka

DoctorEventConsumer

DoctorEventProducer

DoctorService

PatientService

UserService

utils

MedregistryApplication

resources

db.changelog

javadoc

screens

static.css

templates

WHERE

ORDER BY

	id	first_name	last_name	specialization	deleted
8	3	Потап	Иванов	лор	true
9	9	Антон	Чехов	терапевт	true
10	17	Борис	Борисов	диетолог	false
11	18	Иван	Дляудалёнов	физиолог	true
12	10	Модест	Юрьев	травматолог	false
13	20	Иван	Петров	Хирург	false
14	21	Бартоломэо	Бульдогов	Экстрасенс	false
15	22	Захар	Кузьмин	Невролог	false
16	23	Кузьма	Захаров	Окулист	false
17	19	Петр	Иванов	Хирург	false
18	25	Анна2	Смирнова	кардиолог	false
19	24	Анна3	Смирнова	кардиолог	true
20	26	Анна	Смирнова	кардиолог-хирург	false

Run

MedregistryApplication

Console

Actuator

id=?

12:28:05.289 [org.springframework.kafka.KafkaListenerEndpointContainer#0-0-C-1] INFO r.n.m.s.kafka.DoctorEventConsumer - Получено событие об обновлении врача: {id=26, firstName=Анна, lastName=Смирнова, specialization=кардиолог-хирург}

Hibernate:

select

d1_0.id,

d1_0.deleted,

d1_0.first_name,

d1_0.last_name,

d1_0.specialization

from

doctor d1_0

where

Создание пациента:

The screenshot displays an IDE with the following components:

- Project Explorer:** Shows the project structure with folders for logs, src, main, java, ru.natali.medregistry, and config. Files include AppConfig, CacheConfig, and KafkaConfig.
- Code Editor:** Displays the `KafkaConfig.java` file with the following code:

```
12 public class KafkaConfig {  
31     public ObjectMapper objectMapper() {  
32         // ...  
33         mapper.registerModule(new JavaTimeModule());  
34         mapper.disable(SerializationFeature.WRITE_DATES_AS_TIMESTAMPS);  
35         return mapper;  
36     }  
37 }
```
- Run Console:** Shows the output of the application. The output includes logs from Kafka, Spring Framework, and Hibernate. The key log entry is:

```
13:54:02.109 [org.springframework.kafka.KafkaListenerEndpointContainer#1-0-C-1] INFO r.n.m.s.kafka.PatientEventConsumer - Создан пациент: PatientDTO(id=12, firstName=Кирилл, lastName=Степанцов, dateOfBirth=1988-08-08, insuranceNumber=INS987790)
```

Обновление пациента:

The screenshot displays an IDE with the following components:

- Left Panel (Project Explorer):** Shows a project structure with packages like `repository`, `security`, `service`, `impl`, `kafka`, `utils`, and `resources`. The `PatientEventProducer` class is selected under the `kafka` package.
- Center Panel (Code Editor):** Displays the source code of the `PatientEventProducer` class. The code includes a constructor and three methods: `sendPatientCreatedEvent`, `sendPatientUpdatedEvent`, and `sendPatientDeletedEvent`. The `sendPatientUpdatedEvent` method is currently selected, showing its implementation where a `PatientEvent` is created with `eventType: "UPDATE"` and sent to the `"patient-events"` topic.
- Right Panel (Database Explorer):** Shows a connection to `hospitaldb@localhost` with a list of tables including `patient`, `role`, `user_role`, and `users`.
- Bottom Panel (Console):** Displays the output of the application. It shows a log entry from `r.n.m.s.kafka.PatientEventProducer` indicating the successful sending of an update event for a patient with `id=11`, `firstName=Георгий`, `lastName=Степанцов`, and `insuranceNumber=INS987791`. Subsequent logs show the event being received and processed by `PatientEventConsumer`.

Удаление пациента:

DEL localhost:8080/api/m... x +

DELETE  http://localhost:8080/api/medical/patients/14

Params Headers 1 Auth  Body

1

Request DELETE Response 204

▼ HTTP/1.1 204 No Content

Cache-Control	no-cache, no-store, max-age=0, must-revalidate
Connection	close
Date	Wed, 11 Jun 2025 11:19:02 GMT
Expires	0
Pragma	no-cache
X-Content-Type-Options	nosniff
X-Frame-Options	DENY
X-Xss-Protection	0

PatientDTO

UserDTO

exceptions

model

Doctor

DoctorEvent

ERole

Patient

PatientEvent

Role

User

repository

security

service

impl

DoctorServiceImpl

PatientServiceImpl

UserServiceImpl

16 rows

Tx: Auto

DDL

CSV

WHERE

ORDER BY

id	first_name	last_name	date_of_birth	insurance_number	deleted
6	4 Майя	Тожеудалёнова	1990-01-01	INS357851	• true
7	9 Бартоломео	Удалёнов	1998-01-01	INS355009	• true
8	10 Данила	Николаев	1998-01-01	INS355003	false
9	3 Антон	Степанцов	1988-08-08	INS987624	false
10	8 Борис	Бульдогов	1998-01-01	INS355559	• true
11	11 Георгий	Степанцов	1988-08-08	INS987791	• true
12	12 Кирилл	Степанцов	1988-08-08	INS987790	• true
13	13 Степан	Новаязаписев	1998-01-01	INS777888	false
14	15 Степан	Новаязаписев	1998-01-01	INS777890	• true
15	14 Степан	Новаязаписев	1998-01-01	INS777889	• true
16	16 Алекс	Новый	1998-01-01	INS777891	• true

Run MedregistryApplication

Console

Actuator

```

and (
    p1_0.deleted=false
)
Hibernate:
update
patient p1_0
set
deleted=true
where
p1_0.id=?
and (
    p1_0.deleted=false
)
14:27:15.063 [http-nio-8080-exec-7] INFO  r.n.m.s.kafka.PatientEventProducer - Отправка события удаления пациента: PatientEvent{eventType='DELETE', payload=16}
14:27:15.066 [org.springframework.kafka.KafkaListenerEndpointContainer#1-0-C-1] INFO  r.n.m.s.kafka.PatientEventConsumer - Получено событие типа: DELETE
14:27:15.066 [org.springframework.kafka.KafkaListenerEndpointContainer#1-0-C-1] INFO  r.n.m.s.kafka.PatientEventConsumer - Удален пациент с ID: 16

```


DELETE  http://localhost:8080/api/medical/patients/16

Params Headers 1 Auth  Body 

```
1 ▼ {  
2   "firstName": "Алекс",  
3   "lastName": "Новый",  
4   "dateOfBirth": "1998-01-01",  
5   "insuranceNumber": "INS777891"  
6 }
```

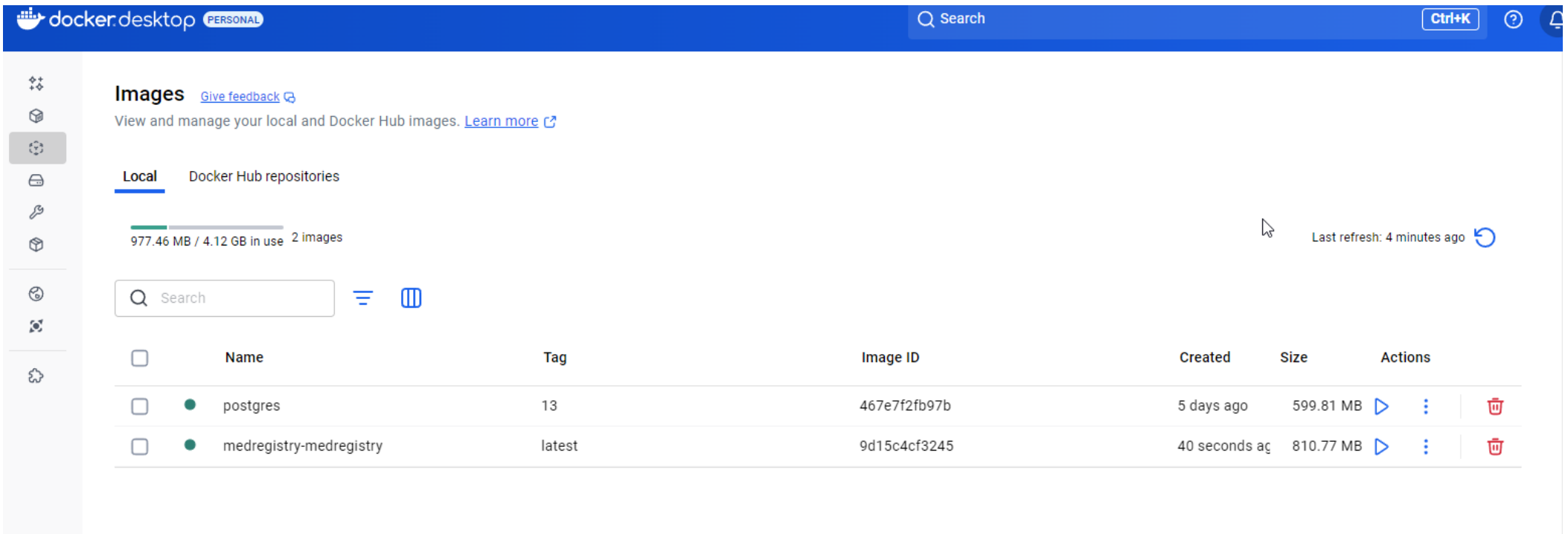
</>

Request DELETE Response 204

▼ HTTP/1.1 204 No Content

Cache-Control	no-cache, no-store, max-age=0, must-revalidate
Connection	close
Date	Wed, 11 Jun 2025 11:27:15 GMT
Expires	0
Pragma	no-cache
X-Content-Type-Options	nosniff
X-Frame-Options	DENY
X-Xss-Protection	0

Образы в Докере:



Скрин работы Докера в IntelliJ Idea:

Project ▾
Dockerfile
application.properties
docker-compose.yml ×
Database

Services
+ -
API HTTP Request
 (GET) requestDoctors | Get all doctors
docker-default
 > Docker-compose: medregistry healthy
 Dev Containers
 Containers
 > Images
 > Networks
 > Volumes
docker-desktop-linux
 > Docker-compose: medregistry healthy
 Dev Containers
 Containers
 > Images
 > Networks
 > Volumes

Console Dashboard

Docker-compose: medregistry | Running: (2/2)
Restart Stop Down

- Container postgres	Waiting	5.8s
[+] Running 3/4dregistry	Created	0.1s
✓ medregistry	Built	0.0s
✓ Network medregistry_medregistry-net	Created	0.0s
- Container postgres	Waiting	5.9s
[+] Running 3/4dregistry	Created	0.1s
✓ medregistry	Built	0.0s
✓ Network medregistry_medregistry-net	Created	0.0s
- Container postgres	Waiting	6.0s
[+] Running 3/4dregistry	Created	0.1s
✓ medregistry	Built	0.0s
✓ Network medregistry_medregistry-net	Created	0.0s
- Container postgres	Waiting	6.1s
[+] Running 3/4dregistry	Created	0.1s
✓ medregistry	Built	0.0s
✓ Network medregistry_medregistry-net	Created	0.0s
- Container postgres	Waiting	6.2s
[+] Running 3/4dregistry	Created	0.1s
✓ medregistry	Built	0.0s
✓ Network medregistry_medregistry-net	Created	0.0s
✓ Container postgres	Healthy	6.2s
[+] Running 3/4dregistry	Starting	6.1s
✓ medregistry	Built	0.0s
✓ Network medregistry_medregistry-net	Created	0.0s
✓ Container postgres	Healthy	6.2s
[+] Running 4/4dregistry	Starting	6.2s
✓ medregistry	Built	0.0s
✓ Network medregistry_medregistry-net	Created	0.0s
✓ Container postgres	Healthy	6.2s
✓ Container medregistry	Started	6.3s

Проверены все тесты с учетом измененных методов для Kafka:

The screenshot displays an IDE window with a project named 'medregistry in medregistry'. The 'Run' tab is active, showing a list of tests on the left and their execution details on the right.

Test Results:

- ✓ DoctorServiceMockTest (171 ms)
 - ✓ deleteDoctor_ShouldSoftDelete (118 ms)
 - ✓ getDoctorById_WithValidId_ShouldReturn (17 ms)
 - ✓ deleteDoctor_ShouldCallSoftDelete (4 ms)
 - ✓ updateDoctor_WithInvalidId_ShouldReturn (3 ms)
 - ✓ deleteDoctor_ShouldThrowException (2 ms)
 - ✓ convertToEntity_ShouldReturnCorrect (5 ms)
 - ✓ deleteDoctor_WithInvalidId_ShouldReturn (2 ms)
 - ✓ createDoctor_ShouldSaveAndReturn (3 ms)
 - ✓ updateDoctor_WithValidId_ShouldReturn (3 ms)
 - ✓ getAllDoctors_ShouldReturnAllDoctors (6 ms)
 - ✓ convertToDto_ShouldReturnCorrect (3 ms)
 - ✓ getDoctorById_ShouldReturnDoctor (2 ms)
 - ✓ getDoctorById_WithInvalidId_ShouldReturn (1 ms)
 - ✓ updateDoctor_ShouldThrowException (2 ms)
- ✓ DoctorControllerTest (71 ms)
 - ✓ updateDoctor_shouldReturnUpdated (62 ms)
 - ✓ getDoctorById_shouldReturnDoctor (2 ms)
 - ✓ getDoctorById_notFound_shouldThrow (2 ms)
 - ✓ deleteDoctor_shouldReturnNoContent (2 ms)
 - ✓ getAllDoctors_shouldReturnListOfDoctors (2 ms)
 - ✓ createDoctor_shouldReturnCreated (1 ms)
- ✓ MedicalApiControllerIntegrationTest (148 ms)
 - ✓ deletePatient_shouldReturn204 (63 ms)
 - ✓ createPatient_withInvalidDate_shouldReturn (37 ms)
 - ✓ updatePatient_shouldReturn200 (20 ms)
 - ✓ getPatientById_shouldReturn200 (7 ms)
 - ✓ createPatient_shouldReturn201 (9 ms)
 - ✓ getPatientByInsuranceNumber_shouldReturn (12 ms)
- ✓ MedicalApiControllerTest (8 ms)
- ✓ PatientServiceImplTest (47 ms)
- ✓ DoctorControllerTest (27 ms)
 - ✓ getDoctorById_NonExistentId_Returns (21 ms)
 - ✓ deleteDoctor_NonExistentId_Returns (6 ms)

Log Output:

```
groupId=medregistry-group] Request joining group due to: consumer pro-actively leaving the group
15:38:30.699 [org.springframework.kafka.KafkaListenerEndpointContainer#1-0-C-1] INFO o.a.k.clients.FetchSessionHandler - [Consumer
clientId=consumer-medregistry-group-12, groupId=medregistry-group] Node 0 sent an invalid full fetch response with extraIds=(EVHB3ARxRYaPqe8YWhvwXw), response=()
15:38:30.699 [org.springframework.kafka.KafkaListenerEndpointContainer#0-0-C-1] INFO o.a.k.clients.FetchSessionHandler - [Consumer
clientId=consumer-medregistry-group-11, groupId=medregistry-group] Node 0 sent an invalid full fetch response with extraIds=(ubjDCMT7Shal3VDsXPxD_g), response=()
15:38:30.700 [org.springframework.kafka.KafkaListenerEndpointContainer#0-0-C-1] INFO o.a.kafka.common.metrics.Metrics - Metrics scheduler closed
15:38:30.700 [org.springframework.kafka.KafkaListenerEndpointContainer#1-0-C-1] INFO o.a.kafka.common.metrics.Metrics - Metrics scheduler closed
15:38:30.700 [org.springframework.kafka.KafkaListenerEndpointContainer#0-0-C-1] INFO o.a.kafka.common.metrics.Metrics - Closing reporter org.apache.kafka.common.metrics
.JmxReporter
15:38:30.700 [org.springframework.kafka.KafkaListenerEndpointContainer#1-0-C-1] INFO o.a.kafka.common.metrics.Metrics - Closing reporter org.apache.kafka.common
.JmxReporter
15:38:30.700 [org.springframework.kafka.KafkaListenerEndpointContainer#0-0-C-1] INFO o.a.kafka.common.metrics.Metrics - Closing reporter org.apache.kafka.common
.telemetry.internals.ClientTelemetryReporter
15:38:30.700 [org.springframework.kafka.KafkaListenerEndpointContainer#0-0-C-1] INFO o.a.kafka.common.metrics.Metrics - Metrics reporters closed
15:38:30.700 [org.springframework.kafka.KafkaListenerEndpointContainer#1-0-C-1] INFO o.a.kafka.common.metrics.Metrics - Closing reporter org.apache.kafka.common
.telemetry.internals.ClientTelemetryReporter
15:38:30.700 [org.springframework.kafka.KafkaListenerEndpointContainer#1-0-C-1] INFO o.a.kafka.common.metrics.Metrics - Metrics reporters closed
15:38:30.703 [org.springframework.kafka.KafkaListenerEndpointContainer#0-0-C-1] INFO o.a.kafka.common.utils.AppInfoParser - App info kafka.consumer for
consumer-medregistry-group-11 unregistered
15:38:30.704 [org.springframework.kafka.KafkaListenerEndpointContainer#1-0-C-1] INFO o.a.kafka.common.utils.AppInfoParser - App info kafka.consumer for
consumer-medregistry-group-12 unregistered
15:38:30.704 [org.springframework.kafka.KafkaListenerEndpointContainer#0-0-C-1] INFO o.s.k.l.KafkaMessageListenerContainer - medregistry-group: Consumer stopped
15:38:30.704 [org.springframework.kafka.KafkaListenerEndpointContainer#1-0-C-1] INFO o.s.k.l.KafkaMessageListenerContainer - medregistry-group: Consumer stopped
15:38:30.705 [SpringApplicationShutdownHook] INFO o.s.o.j.LocalContainerEntityManagerFactoryBean - Closing JPA EntityManagerFactory for persistence unit 'default'
Hibernate: drop table if exists doctor cascade
Hibernate: drop table if exists patient cascade
Hibernate: drop table if exists role cascade
Hibernate: drop table if exists user_roles cascade
Hibernate: drop table if exists users cascade
15:38:30.705 [SpringApplicationShutdownHook] INFO com.zaxxer.hikari.HikariDataSource - HikariPool-6 - Shutdown initiated...
15:38:30.708 [SpringApplicationShutdownHook] INFO com.zaxxer.hikari.HikariDataSource - HikariPool-6 - Shutdown completed.

Process finished with exit code 0
```